



Jahangirnagar University
Department of Computer Science and Engineering
2nd Year 1st Semester B.Sc. (Hons.) Final Examination - 2022

Course Title: **Mathematics III**
Time: **3 Hours**

Course No: **CSE-201**
Full Marks: **60**

[Answer any 6 (Six) from the following. All questions are of equal marks. Figures in the right margin indicate marks.]

1. a. Define divergence and curl of a vector field. 2
b. When is a vector said to be solenoidal, and when is it irrotational? Find a constant a, b, c so that $\underline{v} = (x + az)\underline{i} + (ax + by + z)\underline{j} + (4x + cy + 2bz)\underline{k}$ is irrotational. 4
c. Calculate $\underline{\nabla} \cdot \underline{\nabla} \phi$ and $\underline{\nabla} \times \underline{\nabla} \phi$ for the scalar field $\phi = 2x^3y^2z^4$. 4
2. a. Define directional derivative. Find the directional derivative of $\phi = x^2y^2z + 4xz^2$ at $(1, 2, 1)$ in the direction $2\underline{i} + \underline{j} - 2\underline{k}$. 4
b. Show that $\underline{\nabla} \phi$ is a vector perpendicular to the surface $\phi(x, y, z) = c$ where c is a constant. 3
c. Show that $\underline{\nabla} \cdot \left(\frac{\underline{r}}{r^3} \right) = 0$. Here, $\underline{r}$ is the position vector. 3
3. a. What is Argand diagram? Also, define modulus and argument of a complex number. 2
b. What is analytic function? Find the necessary and sufficient condition for a complex function $f(z) = u(x, y) + iv(x, y)$ to be analytic at a point in a region R . 4
c. Prove that $u = e^{-x}(x \sin y - y \cos y)$ is a harmonic function. Then find v such that $f(z) = u + iv$ is analytic. 4
4. a. State and prove the Cauchy integral formula. 4
b. Evaluate $\oint_C \frac{e^{2z}}{(z-1)^4} dz$; where C is the circle $|z| = 3$. 2
c. State and prove Liouville's Theorem. 4

5. a. Find the Laurent series about the indicated singularity of the following functions- 5

i) $\frac{z - \sin z}{z^3}; z = 0$

ii) $\frac{z}{(z+1)(z+2)}; z = -2$

b. Evaluate $\int_0^{\infty} \frac{dx}{x^4 + 81}$ 5

6. a. Find the Laplace transform of the function $(t + 2)^2 e^t$ 4

b. Solve the following differential equations by using Laplace transform 6

i. $\frac{dx}{dt} + 5x = e^{2t}, \quad x(0) = 1$

ii. $\frac{d^2 y}{dt^2} + y = 3 \sin(2t), \quad y(0) = 3, y'(0) = 1.$

7.

Define Fourier series. Find the Fourier series of

$$f(x) = \begin{cases} 0 & ; -\pi \leq x < 0 \\ \frac{\pi x}{4} & ; 0 \leq x \leq \pi \end{cases}$$

8. a) Find the Fourier transform of $F(x) = \begin{cases} 1 & |x| < a \\ 0 & |x| > a \end{cases}$. Also draw the graph of $F(x)$ and its 6

Fourier transform for $a = 1$.

b) Expand $F(x) = \begin{cases} 2 - x & 0 < x < 4 \\ x - 6 & 4 < x < 8 \end{cases}$ in a Fourier series of period 8. 4



Jahangirnagar University
Department of Computer Science and Engineering
Second Year 1st Semester B.Sc. (Honors) Final Examination - 2022

Course Title: Numerical Methods
Time: 3.00 Hours

Course No: CSE-205
Full Marks: 60

[Answer each of the following questions. Each question carries equal marks. Figures in the right margin indicate marks.]

1.

Section-I (CO1)

(Answer all the questions.)

- a) **Define** Numerical Computing. Explain why numerical methods are used to solve engineering problems. 2
- b) **List out** the difference between Gauss elimination and Gauss-Jordan methods of solving simultaneous equations critically. 2
- c) **Define** convergence? How is it important in numerical computing? 2
- d) **State** curve fitting? Why it is needed in numerical computing? 2
- e) **Define:** i) Regression ii) Interpolation 2
- f) **Outline** the difference between initial-value problem and boundary-value problems. 2

2.

Section-II (CO2)

(Answer Any Three out of Four)

- a) **Explain** that $f(x) = x^4 + x - 1$ has a real root α in the interval $[0.5, 1.0]$. 4
 - b) With appropriate figure **explain** the working principle of Bisection method for finding a root of nonlinear equation. 4
 - c) **Illustrate** and apply the least square regression to the following set of data to fit a straight line: 4
- | | | | | | | | |
|----|---|---|---|---|---|---|----|
| x: | 1 | 3 | 4 | 6 | 8 | 9 | 11 |
| y: | 1 | 2 | 4 | 4 | 5 | 7 | 8 |
- d) **Demonstrate** and Derive Newton's divided difference interpolation polynomial. Mention some merits and demerits of this method. 4

3.

Section-III (CO3)

(Answer Any Three out of Four)

- a) **Derive** a formula for computing second derivative of a function using Taylor's expansion. Find approximation to second derivative of $\cos(x)$ at $x=0.75$ with $h=0.01$. Compare with the true value. 4
- b) **Compute** the integral $\int_0^{\pi/2} \sqrt{\sin x} dx$ applying Simpson's 1/3 rule for $n = 6$ with an accuracy to five decimal places. 4

- c) **Demonstrate** Newton's divided difference interpolation polynomial. Also **show** some merits and demerits of this method. 4
- d) **Apply** Euler's method to estimate $y(2)$ using i) $h = 0.5$ and ii) $h = 0.25$ 4

$$\frac{dy}{dx} = 3x^2 + 1, \quad y(1) = 2$$

4.

Section-IV (CO4)

(Answer Any Three out of Four)

- a) **Solve** the following system of equations by Gauss-Seidel iteration method (show at least five iterations): 4

$$x_1 + 2x_2 - 3x_3 = 4$$

$$2x_1 + 4x_2 - 6x_3 = 8$$

$$x_1 - 2x_2 + 5x_3 = 4$$

- b) With appropriate figure **describe** composite trapezoidal rule for computing integrals. 4

- c) **Examine** and fit a power-function model of the form, $y = ax^b$ 4

x	1	2	3	4	5
y	0.5	2	4.5	8	12.5

- d) Consider the following equation: 4

$$\frac{d^2y}{dx^2} = e^{x^2} \quad \text{with } y(0) = 0 \quad \text{and } y(1) = 0$$

Apply and **analyze** Finite Difference Method to estimate the values of $y(x)$ at $x = 0.25, 0.5$, and 0.75 .

5.

Section-V (CO5)

(Answer Any Two out of Three)

- a) The population of a country for various years in million is provided. Using Newton's forward difference interpolation formula, **estimate** the population for the year 1988. 6

Year x :	1981	1991	2001	2011	2021
Population y :	46	66	81	93	101

- b) Determine and **devise** the coefficients of the characteristic polynomial of the system: 6

$$(-1 - \lambda)x_1 = 0$$

$$x_1 + (-2 - \lambda)x_2 + 3x_3 = 0$$

$$2x_2 + (-3 - \lambda)x_3 = 0$$

Using the Faddeev-Leverrier method.

- c) For the following differential equation, **estimate** $y(2)$ using i) Euler's method and ii) Heun's method using $h = 0.25$. Compare the results with exact answers. 6

$$y'(x) = 2y/x \quad \text{with } y(1) = 2.$$



Jahangirnagar University
Department of Computer Science and Engineering
2nd Year 1st Semester B.Sc. (Hons.) Final Examination -2022

Course Title: Computer Ethics and Cyber Law
Time: 3 Hours

Course No: **CSF-203**
Full Marks: **60**

[Answer each of the following questions. Each question carries equal marks. Figures in the right margin indicate marks.]

1. Question No. 1 will be based on **CO1**. This question will be consisting of **Six** Sections. Each section may contain several subsection like (a (i), a(ii) etc)

Students have to answer *All of them*.

- a) **What** is a profession? **How** is a computer-related career unlike a fully developed profession? 2
- b) Despite the increasing popularity of open-source software, **why** do people believe that proprietary software is more reliable to use? 2
- c) Work from home is a new trend nowadays, where companies are allowing their employees to use their own laptops and submit work online from anywhere. **Discuss** how this increases productivity. 2
- d) If a person commits a cyber-crime outside Bangladesh, such as fabricating a record, does the amended definition of 'evidence' and 'fabrication of record' will make him/her liable under the IT Act? 2
- e) **What** is a peer-to-peer network? **How** can it be used to violate intellectual property rights? 2
- f) **What** constitutes evidence and **how** electronic evidence is different from paper evidence? 2
- g) Proponents of globalization claim that it helps workers in developing countries. Opponents of globalization claim the opposite. Summarize the arguments pro and con. 2

2. Question No. 2 will be based on **CO2**. This question will be consisting of **Five** Sections. Each section may contain several subsections like (a (i), a(ii), a(iii) etc)

Students have to answer *Any Four* out of *Five*.

- a) "Ethics is a set of rules for determining moral standards or socially acceptable behavior"- do you agree with the statement? **Elaborate** your answer with some examples. 3
- b) **What** is Corporate Social Responsibility (CSR)? **Why** is fostering good business ethics important for IT industry? 3
- c) In today's computing environment of increasing complexity, higher user expectations, expanding and changing systems, and increased reliance on software with known vulnerabilities, it is no wonder that the number, variety, and impact of security incidents are increasing dramatically. Considering the above statement, explain **why** computer incidents are so prevalent. 3
- d) Answer the followings: 3
 - (i) Can you think of any scenario where hacking may be advantageous or useful? Discuss with examples. [2 marks]
 - (ii) Is it morally acceptable to use a denial-of-service attack to shut down a Web server that distributes child pornography? Give reasons behind your opinion. [1 mark]
- e) Answer the followings: 3

(i) What is data mining? How could "cookies" created by a Web server affect a computer user's privacy? [2 marks]

(ii) Suppose company A wants to develop a program that duplicates the functionality of a program made by company B. Describe how company A may do this without violating the copyrights held by company B. [1 mark]

3. Question No. 3 will be based on **CO3**. This question will be consisting of **Four** Sections. Each section may contain several subsections like (a (i), a(ii), a(iii) etc)

Students have to answer **Any Three** out of **Four**.

- a) **Differentiate** digital signature with electronic signature. What is the main purpose of the digital signature? Write down the importance of digital signature in the all formal electronic issuance of digital documents in Jahangirnagar University. 4
- b) **Describe** similarities and differences between subjective relativism and ethical egoism. 4
- c) **Compare** among the following terms: (i) Copyright (ii) Patent and (iii) Trade secret. 4
- d) What is data encryption? **Differentiate** between *secret/symmetric-key encryption* and public-key encryption. 4

4. Question No. 4 will be based on **CO4**. This question will be consisting of **Four** Sections. Each section may contain several subsections like (a (i), a(ii), a(iii) etc)

Students have to answer **Any Three** out of **Four**.

- a) "You are the vice president for software development at a small, private firm. Sales of your firm's products have been strong, but you recently detected a patent infringement by one of your larger competitors. Your in-house legal staff has identified three options: (1) Ignore the infringement out of fear that your larger competitor will file numerous countersuits; (2) threaten to file suit, but try to negotiate an out-of-court settlement for an amount of money that you feel your larger competitor would readily pay; or (3) point out the infringement and negotiate aggressively for a cross-licensing agreement with the competitor, which has numerous patents you had considered licensing." 4

Use the five-step decision-making process to **analyze** the above situation and **recommend** a course of action. Which option would you pursue and why? And why do you not **choose** the other options?

- b) "You are a recent college graduate with only a year of experience with your employer. You were recently promoted to manager of email services. You are quite surprised to receive a phone call at home on a Saturday from the Chief Financial Officer of the firm asking that you immediately delete all email from all email servers, including the archive and back-up servers, that is older than six months. He states that the reason for his request is that there have been an increasing number of complaints about the slowness of email services. In addition, he says he is concerned about the cost of storing so much email. This does not sound right to you because you recently have taken several measures that have speeded up email services. An alarm goes off when you recall muted conversations in the lunchroom last week about an officer of the company passing along insider trader information to an executive at a hedge fund." 4

What do you say to the Chief Financial Officer? Use the five-step decision-making process to **analyze** the above situation and **recommend** a course of action.

- c) *"You are the customer support manager for a small software manufacturer. The newest addition to your 10-person team is Sofia, a recent college graduate. She is a little overwhelmed by the volume of calls, but is learning quickly and doing her best to keep up. Today, as you performed your monthly review of employee email, you were surprised to see that Sofia has received several messages from employment agencies. One message says, 'Sofia, I'm sorry you don't like your new job. We have lots of opportunities that I think would much better match your interests. Please call me, and let's talk further.' You're shocked and alarmed. You had no idea she was unhappy, and your team desperately needs her help to handle the onslaught of calls generated by the newest release of software. If you're going to lose her, you'll need to find a replacement quickly. You know that Sofia did not intend for you to see the email, but you can't ignore what you saw."* 4

Should you confront Sofia and demand to know her intentions? Should you avoid any confrontation and simply begin seeking her replacement? Could you be misinterpreting the email? What should you do? Consider the software code of ethics to **analyze** the situation and **recommend** a course of action.

- d) **Explain** the ethical relationship between IT professionals and Clients. Consider the following Scenario: 4

"You are a new brand manager for a product line of Coach purses. You are considering purchasing customer data from a company that sells a large variety of women's products online. In addition to providing a list of names, mailing addresses, and email addresses, the data includes an estimate of customers' annual income based on the zip code in which they live, census data, and highest level of education achieved. You could use the data to identify likely purchasers of your high-end purses, and you could then send those people emails announcing the new product line and touting its many features."

Analyze the advantages and disadvantages of such a marketing strategy. Would you recommend this means of promotion in this instance? Why or why not?

5. Question No. 5 will be based on **CO5**. This question will be consisting of **Three** Sections. Each section may contain several subsections like (a (i), a(ii), a(iii) etc)
Students have to answer **Any Two** out of **Three**.

- a) Consider a cybercrime scenario as below: 6

"You are the senior software engineer at a start-up company developing an exciting new product that will allow salespeople to generate and email sales quotes and customer invoices from their smartphones."

Your company's sales force has led a major corporation to believe your product will be available next week. Unfortunately, at this point the software still contains quite a few bugs. The leader of the testing group has reported that all of the known bugs appear to be minor, but it will take another month of testing for his team to be confident the product contains no catastrophic errors."

Because of the fierce competition in the smartphone software industry, it is critical that your company be "first to market." To the best of your knowledge, a well-established company will release a similar product in a few weeks. If its product appears first, your start-up company will probably go out of business."

Considering the above scenario, answer the followings:

- (i) Should you recommend release of the product next week? [1 mark]
- (ii) **Judge** your opinion in (i) using one or more appropriate ethical theory. [2 marks]
- (iii) Consider the software code of ethics to **analyze** the situation and **recommend** a course of action. [3 marks]

- b) *In the US, the largest ever case of identity theft occurred when an employee at a software company sold consumer data to a network of criminals. The passwords and codes enabled the gang to download thousands of credit reports – and to cause losses of between US\$ 50 to 100 million by using the information to perform financial transactions. But identity theft isn't always an inside job: In 2009, hackers took advantage of security weak spots to steal 45.5 million debit card numbers from a payment processing company. Within just twelve hours, they had withdrawn more than US\$9.4 million from around 2,100 cash machines in 280 countries worldwide.* 6

Answer the followings:

- i) Following the above story, **create** a real-like cybercrime scenario in the context of Bangladesh relating identity theft. [2 marks]
- ii) **Judge** the scenario in (i) considering one or more appropriate ethical theory. [2 marks]
- iii) Do you think the current Digital Security Act in Bangladesh will cover the cybercrime in (i)? If so, **identify** the clauses are in consideration for this crime and the corresponding offences and punishments. [2 marks]

- c) *One of the leading cybercrime cases is the Bank ABC case where a management trainee of the bank was engaged to be married. The couple exchanged many emails using the company computers. After some time the two broke up and the girl created fraudulent email ids such as "banglabarassociations" and sent emails to the boy's foreign clients. She used the bank's computer to do this. The boy's company lost a large number of clients and took the bank to court.* 6

Answer the followings:

- (i) What are the major laws to be considered in the typical cyber law? [2 marks]
- (ii) According to Bangladesh Cyber Law, what would be the punishment you would proceed with? State the rules of the Law you should follow for the judgment. [4 marks]

----- End of question script -----



Jahangirnagar University
Department of Computer Science and Engineering
2nd Year 1st Semester B.Sc. (Hons.) Final Examination -2022

Course Title: Electronic Devices and Circuit-II
Time: 3 hrs

Course No: **CSE-207**
Full Marks: **60**

[Answer each of the following questions. Each question carries equal marks. Figures in the right margin indicate marks.]

1. Answer *All of them*

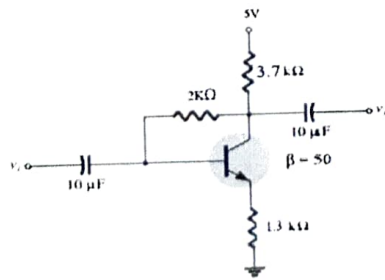
- | | | |
|----|--|---|
| a) | State the meaning of virtual ground. | 2 |
| b) | Mention the working principle of voltage buffer. | 2 |
| c) | How can we find the cutoff frequency? | 2 |
| d) | List the basic features of Op-Amp. | 2 |
| e) | Mention the applications of photodiode. | 2 |
| f) | Sketch the terminal identification of SCR. | 2 |

2. Answer *Any Three* out of *Four*.

- | | | |
|----|--|---|
| a) | Discuss the following questions: | 2 |
| | i. Differentiate between Hartley and Colpitts oscillator. | 2 |
| | ii. Which type of material is used for infrared LED? | |
| b) | Formalize the characteristics and applications of Schottky Diode. | 4 |
| c) | Discuss the function of using an inverting constant-gain multiplier. Draw the diagram. | 4 |
| d) | Explain the basic Instrumentation amplifier model with proper figure. | 4 |

3. Answer *Any Three* out of *Four*.

- | | | |
|----|--|---|
| a) | Determine the voltage gain, input, and output impedance with feedback. For the voltage-series feedback having $A = -50$, $R_i = 5 \text{ k}\Omega$, and $R_o = 10 \text{ k}\Omega$ for feedback $B = -0.10$ Determine: | 4 |
| | i. Voltage gain | |
| | ii. Input impedance | |
| | iii. Output impedance | |
| b) | Calculate the cutoff frequency of an op-amp having specified values
$B_1 = 2 \text{ MHz}$ and $AVD = 200 \text{ V/mV}$. | 4 |
| c) | Determine the quiescent levels of I_{CQ} and V_{CEQ} for the configuration using the exact and approximate techniques. | 4 |



d) Answer the following questions:

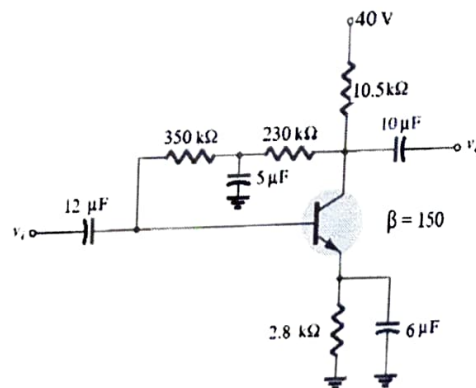
- i. Does UJT work on AC or DC?
- ii. Which type of material is the channel of a unijunction transistor made up of?
 - I. PN type
 - II. It doesn't affect the working
 - III. P type
 - IV. N type

Answer **Any Three** out of **Four**.

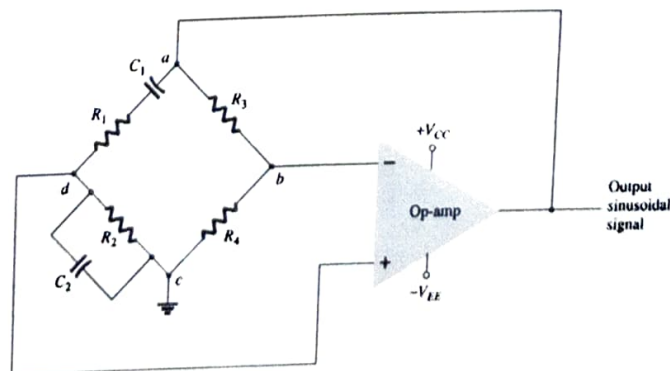
4. a) Answer the following questions:

- i. How to make a Darlington pair transistor?
- ii. For the Voltage feedback network, determine:

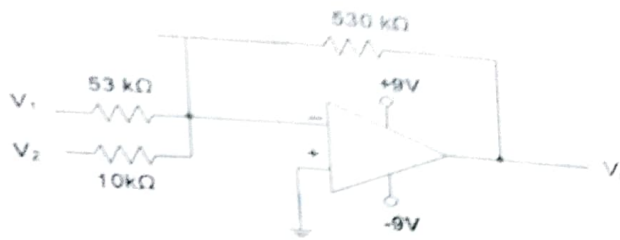
I. I_C	II. V_C
III. V_E	IV. V_{CE}



- b) For the Wien Bridge Oscillator circuit, find out the frequency where $R_1=2K\Omega$, $R_2=4K\Omega$, $R_3=1K\Omega$, $R_4=1K\Omega$, $C_1=1\mu F$, $C_2=2\mu F$.



- c) Calculate the output voltage for the circuit when the inputs are $V_1 = 50 \text{ mV} \sin(1000 t)$ and $V_2 = 10 \text{ mV} \sin(3000 t)$.



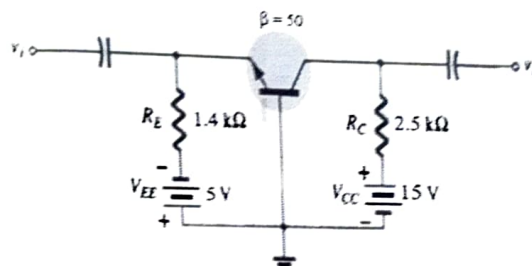
d) Answer the following:

- Analyze the comparison between SCR and GTO.
- Draw the schematic symbol of DIAC.

2
2

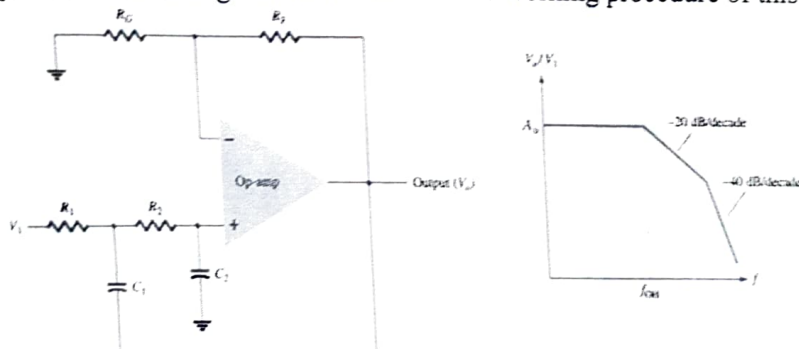
Answer *Any Two* out of *Three*.

- a) Determine the currents I_E and I_B and the voltage V_{CE} and V_{CB} for the common-base configuration. 6



- b)
 - Differentiate between class-C and class-D amplifier.
 - Which type of filter circuit given here? Describe the working procedure of this filter.

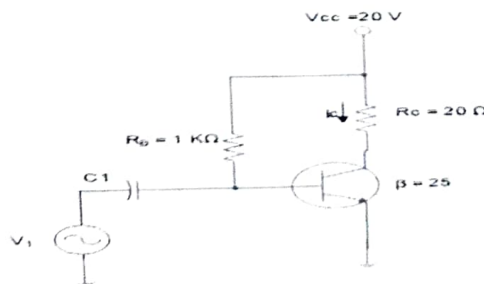
2
4



c) Answer the following:

- Specify the input offset voltage.
- Calculate the Input power, Output power, and Efficiency of the amplifier circuit where an input voltage that results in a base current of 20 mA peak.

2
4





Jahangirnagar University
Department of Computer Science and Engineering
2nd Year 1st Semester B.Sc. (Hons.) Final Examination -2022

Course Title: Algorithms-I
Time: 3 Hours

Course No: CSE 209
Full Marks: 60

Read the question carefully. There are five sections. Each section carries equal marks.
Figures in the right margin indicate marks.

1)

SECTION-1

Section No. 1 is based on **CO1**. There are seven questions.
You have to answer **any six questions** from this section.

- a) Mention two differences between Greedy strategy and Dynamic Programming. 2
- b) Recognize the complexity for the following algorithms: 2
- | Algorithm | Best Case Complexity | Worst Case Complexity |
|----------------|----------------------|-----------------------|
| Linear Search | $\Omega(1)$ | $O(N)$ |
| Insertion sort | | |
| Quick Sort | | |
| Heap Sort | | |
- c) List out the properties of a binary search tree. 2
- d) Explain the divide and conquer paradigm. 2
- e) Recall the properties of a dynamic programming algorithm. 2
- f) Proof the complexity of the Dijkstra algorithm is $O((V + E) \log_2 V)$. 2
- g) Explain tree edge, forward edge, cross edge, and back edge. 2

2)

SECTION-2

Section No. 2 is based on **CO2**. There are four questions.
You have to answer **any three questions out of four** from this section.

- a) Explain the Merge sort algorithm with the following example. 2+2
 $\{8, 5, 7, 1, 6, 4, 3\}$
Hence, explain the complexity merge sort is $O(n \log n)$ recursion tree method.
- b) Explain the Quick sort with the following sequence as example (step by step) always considering the leftmost element of an array/sub array as the pivot element. You have to show the simulation for the first iteration only. 2+2
 $\{3, 5, 2, 4, 7, 6, 1\}$

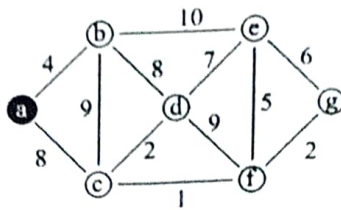
If you have two options, (1) quick sort and (2) randomized quick sort, which one you will choose and why- explain briefly.

- c) Consider the above algorithm and answer the following: 4

ALGORITHM *Enigma*($A[0..n-1, 0..n-1]$)
//Input: A matrix $A[0..n-1, 0..n-1]$ of real numbers
for $i \leftarrow 0$ to $n-2$ do
 for $j \leftarrow i+1$ to $n-1$ do
 if $A[i, j] \neq A[j, i]$
 return false
return true

Answer the following:

- i. What is its basic operation?
- ii. How many times is the basic operation executed?
- d) Consider "**a**" as the source node. Identify the appropriate algorithm to find the single source shortest path from node **a**. 4



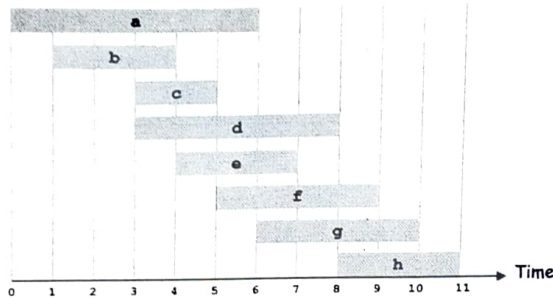
SECTION-3

Section No. 3 is based on **CO3**. There are four questions.

You have to answer **any three questions out of four** from this section.

a)

4



Given the schedule of different lectures of a department for a particular day. Each lecture has a start time and a finish time.

Identify optimal scheduling strategy to minimize the total number of classrooms to schedule all lectures so that no two occur at the same time in the same room. Demonstrate the solution briefly.

- b) Illustrate a binary search tree for the following sequence. 4

$$S = \{10, 22, 9, 33, 21, 50, 41, 60, 80\}$$

- c) Compute the Longest Common Subsequence (LCS) of the strings $X = \text{"ABCBDAB"}$ and $Y = \text{"BDCABA"}$ using dynamic programming. Show how you calculated the LCS using a matrix containing both the cost and the arrows for each cell in the matrix. 4

- d) Apply CRT (Chinese Remainder Theorem) to solve the simultaneous congruences, 4
 $x \equiv 6 \pmod{11}, x \equiv 13 \pmod{16}, x \equiv 9 \pmod{21}, x \equiv 19 \pmod{25}.$

SECTION-4

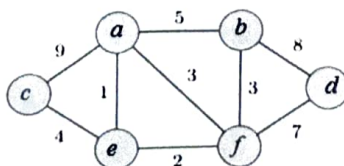
Section No. 4 is based on **CO4**. There are four questions.

You have to answer **any three questions out of four** from this section.

- 4) a) Analyze the application for Huffman coding in the field of computer science within two sentences. Hence, point out the length of the codeword using Huffman coding. 1+3

	a	b	c	d	e	F
Frequency (thousands)	34	14	12	16	14	10

- b) Examine the MST for the above graph using Prim's or Kruskal's algorithm. 4
 Distinguish between Prim's and Kruskal's algorithm for finding the MST in terms of: (i) Time Complexity and (ii) Preferred Graph.



- c) Compute the optimal price for the 0-1 knapsack problem for knapsack capacity, $W=8$. Which items will be in the optimal solution? What will be their total price? Analyze the optimal solution with appropriate tabular illustration using a matrix containing both price and arrows for each cell in the matrix. 4

item	A	B	C	D	E	F
Weight	2	1	5	2	4	3
price	7	8	14	5	10	15

- d) A list of different activities with starting and ending times. Calculate the maximum size of non-overlapping activities that can be performed. 4

Activities	1	2	3	4	5	6	7	8
Start Time	1	0	1	4	2	5	3	4
Finish Time	3	4	2	6	9	8	5	5

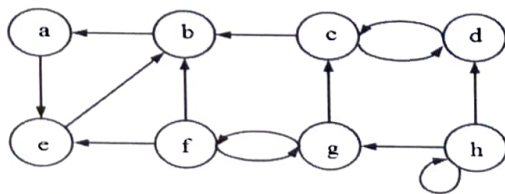
Analyze the complexity of the solution using big O notation.

SECTION-5

Section No. 5 is based on CO5. There are three questions.

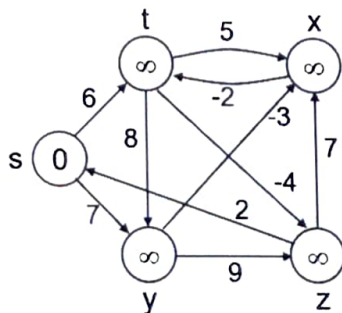
You have to answer **any two questions out of three** from this section.

- 5) a) Prepare a list of key steps for finding SCC (Strongly Connected Component) in a given graph. Hence, find SCC for the given graph using Kosaraju Algorithm illustrating all the steps. 2+3+1



Evaluate the complexity of Kosaraju algorithm.

- b) "If you have a cycle in the graph, that has in sum a negative weight, then there is no shortest path."-justify your answer. From the following graph below, find whether a negative weight cycle exists or not using the Bellman ford algorithm. 5+1

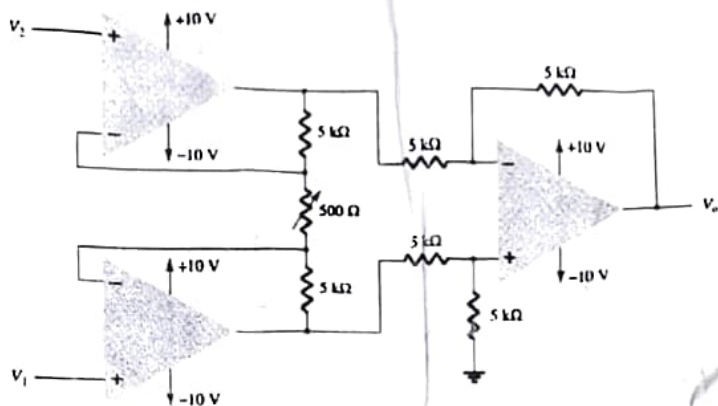


Hence, evaluate the complexity of the algorithm.

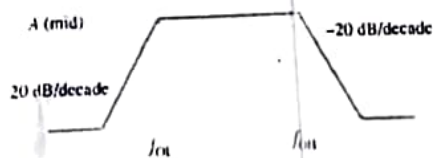
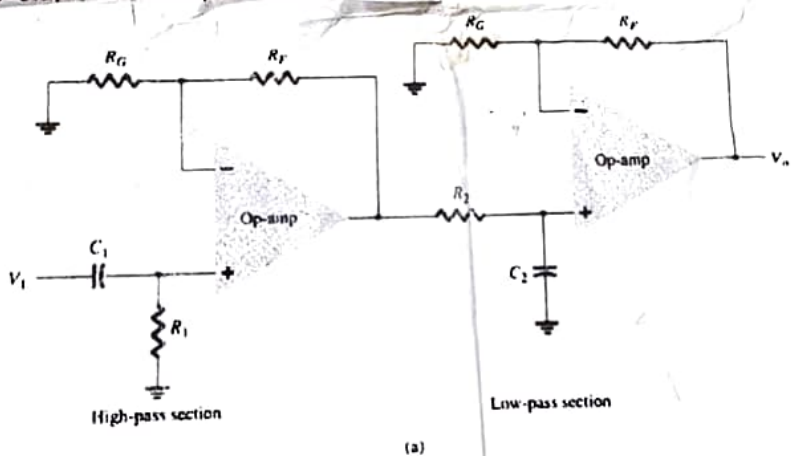
- c) Consider the iterative version of the RADIX SORT Algorithm. Assume that we are given an array of n nonnegative integers whose values range between 0 and 2^n . 3+3
- What is the time complexity of the algorithm in terms of the number of digit comparisons, assuming that the array contains n integers? Justify your answer.
 - In the case above, is it better to use SELECTION SORT or RADIX SORT? Justify your answer.

-----0-----
Good Luck!!!!

1. Mention the activities of zero drift amplifier.
2. State the slew rate.
3. Demonstrate the characteristics of voltage buffer.
4. Calculate the output voltage expression for the circuit.

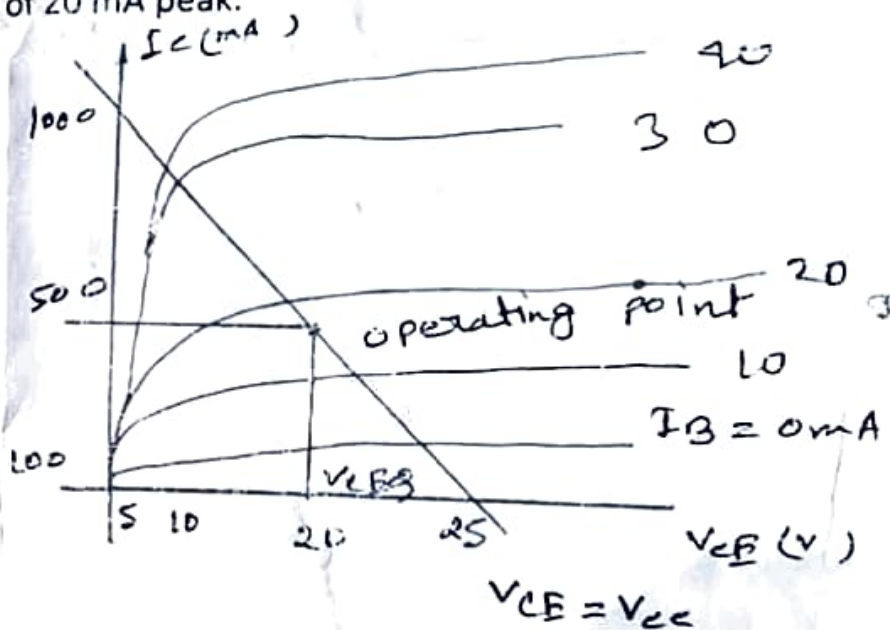
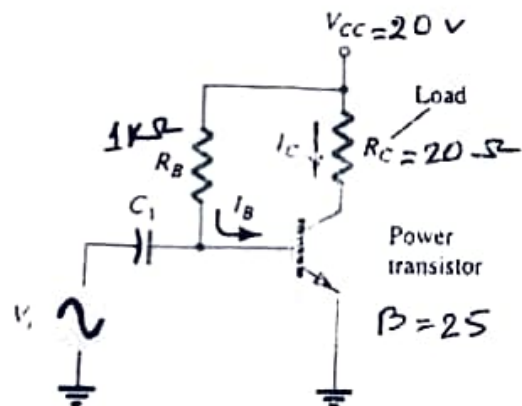


5. Calculate the cut-off frequency of the bandpass filter where $R_1=R_2=10\text{K}\Omega$, $C_1=0.1\mu\text{F}$, $C_2=0.002\mu\text{F}$.



Jahangirnagar University
Department of Computer Science and Engineering
2nd Year 1st Semester CT-2
Course Code: CSE-207
Course Name: Electronic Device and Circuit-II
Time: 30 min Marks:10

1. State the Amplifier Distortion. 2
2. Draw and describe about the Transformer-Coupled Push-Pull Class B Amplifier. 3
3. Calculate the Input power and efficiency of the amplifier circuit for an input voltage that results in a base current of 20 mA peak. 5



JAHANGIRNAGAR UNIVERSITY

Department of Computer Science and Engineering

Course Code: 207 CT-3

Course Title: Electronics Device and circuit theory -II

Time: 20min

Full Marks: 10

5. Draw and describe the operation, characteristics and applications of SCR.
6. If $V_{BB}=10V$, $R=20K\Omega$, $C=1\mu F$, $R_K=100\Omega$, $R_{B1}=10K\Omega$, $R_{B2}=5K\Omega$, $I_P=100\mu A$, $V_V=1V$, $I_V=5mA$, determine:
 - g. $V_P=?$
 - h. $R_{max}, R_{min}=?$
 - i. T and frequency of oscillation.

CT #1

Answer any 3 questions.

Time: 30 minutes

Full Marks: 15

1. What is numerical computing? Discuss various phases of numerical computing.
2. Discuss different types of errors encountered in numerical computing.
3. Explain Bisection method for finding a root of nonlinear equation. Mention some merits and demerits of this method.
4. Derive the secant formula. How is it different from the false position formula?

Answer any 2 questions.

Time: 30 minutes

Marks: 20

1. Describe composite trapezoidal rule for computing integrals.
2. Derive a formula for computing second derivative of a function using Taylor's expansion. Find approximation to second derivative of $\cos(x)$ at $x=0.75$ with $h=0.01$. Compare with the true value.

3. Compute the integral $\int_0^{\pi/2} \sqrt{\sin x} dx$ applying Simpson's 1/3 rule for $n = 6$ with an accuracy to five decimal places.

Answer any ONE question. Time: 15 minutes Marks: 10

1. Determine the coefficients of the characteristic polynomial of the system:

$$(-1 - \lambda) x_1 = 0$$

$$x_1 + (-2 - \lambda) x_2 + 3x_3 = 0$$

$$2x_2 + (-3 - \lambda) x_3 = 0$$

Using the Fadeev-Leverrier method.

2. For the following differential equation, estimate $y(2)$ using i) Euler's method and ii) Heun's method using $h = 0.25$. Compare the results with exact answers.

$$y'(x) = 2y/x \quad \text{with } y(1) = 2.$$

3. Consider the following equation:

$$\frac{d^2 y}{dx^2} = e^{x^2} \quad \text{with } y(0) = 0 \quad \text{and } y(1) = 0$$

Apply Finite Difference Method to estimate the values of $y(x)$ at $x = 0.25, 0.5$, and 0.75 .

Any 2 questions. Time: 30 minutes Marks: 20

1. Define curve fitting. Derive Newton's divided difference interpolation polynomial. Mention some merits and demerits of this method.

2. Obtain the solution of the following system using Gauss-Seidel iteration method:

$$2x - y + 2z = 6$$

$$2x - y + z = 3$$

$$x + 3y - z = 4$$

3. Use the least square regression to fit a straight line to the following set of data.

x	1	3	4	6	8	9	11
y	1	2	4	4	5	7	8

CSE 203 Computer Ethics and Cyber Law – Tutorial 1 Mark 20 (5x4)

Q1. For the following offence, (i) give an example and detail scenario (create a story) of the offence (ii) punishment of the offence under Digital Security Act–

“E-transaction without legal authority”

Q2. Compare among the following terms: (i) Morals (ii) Ethics (iii) Law (iv) Virtue (v) vice.

Q3. People give a variety of reasons for copying a Movie DVD from a friend instead of buying it. Refute each of the reasons given below, using one of the viable ethical theories described in your class. (You don't have to use the same theory each time.)

a. I'm giving my friend the opportunity to do a good deed.

b. Everyone else is doing it. Why should I be the only person to buy it when everyone else is getting it for free?

Q4. What is corporate social responsibility, and why is fostering good business ethics important?

CSE 203 Computer Ethics and Cyber Law – Tutorial 1 Mark 20 (5x4)

Q1. Explain the ethical relationship between IT professionals and Clients.

Q2. Consider, you are an information security consultant of a government organization that stores sensitive public information. Suggest few steps to be taken to reduce vulnerabilities to cyber-attacks.

Q3. In many countries, although commoners have to go through stringent checks at airports and other important checkpoints, VIPs have special privileges and are allowed to bypass security verifications. Do you think this is lawful and ethical? How about privacy issue?

Q4. Compare among the following terms: (i) Copyright (ii) Patent and (iii) Trade secret.

Q1. What is the digital divide? Where does it exist? Why is it important to bridge the digital divide? [5 marks]

Q2. Proponents of globalization claim that it helps workers in developing countries. Opponents of globalization claim the opposite. Summarize the arguments pro and con. [5 marks]

Q3. *"You are the customer support manager for a small software manufacturer. The newest addition to your 10-person team is Sofia, a recent college graduate. She is a little overwhelmed by the volume of calls, but is learning quickly and doing her best to keep up. Today, as you performed your monthly review of employee email, you were surprised to see that Sofia has received several messages from employment agencies. One message says, "Sofia, I'm sorry you don't like your new job. We have lots of opportunities that I think would much better match your interests. Please call me, and let's talk further." You're shocked and alarmed. You had no idea she was unhappy, and your team desperately needs her help to handle the onslaught of calls generated by the newest release of software. If you're going to lose her, you'll need to find a replacement quickly. You know that Sofia did not intend for you to see the email, but you can't ignore what you saw."*

Should you confront Sofia and demand to know her intentions? Should you avoid any confrontation and simply begin seeking her replacement? Could you be

misinterpreting the email? What should you do? Consider the software code of ethics to analyze the situation and recommend a course of action. [10 marks]

Your Name:

Your Class Roll:

Answer all the questions. Each question Carries Equal Marks. So, the total mark is, $6 \times 5 = 30$

1) $T(n) = 2T(n/2) + cn$

- **Proof the complexity of the solution using the substitution method.**

$$2) \quad T(n) = T(n/2) + c$$

- **Proof the complexity of the solution using the recursion tree method.**

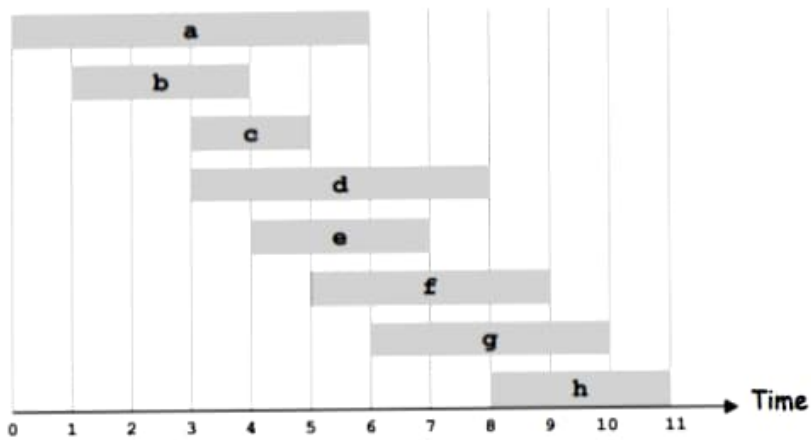
3) State Master Method

-

$$T(n) = 2T(n/2) + n \log n$$

- **Proof the complexity of the solution using the master method.**

4)



Given the schedule of different lectures of a department for a particular day. Each lecture has a start time and a finish time. Find the minimum number of classrooms to schedule all lectures so that no two occur at the same time in the same room.

- 5) Sort the following sequence using merge sort algorithm.
{4, 9, 1, 8, 5, 6, 2, 3, 7}

Your Class Roll:

Answer all the questions. Each question Carries Equal Marks. So, the total mark is, $5 \times 6 = 30$

- 1) $X = \text{"GATTACA"}$ and $Y = \text{"GCATGAC"}$ Find the LCS between X and Y .
- 2) Find the longest increasing subsequence from the given sequences using LIS (N^2 Complexity)
 $\{3, 1, 4, 7, 6, 8, 9, 10, 6\}$
- 3) Find the longest increasing subsequence from the given sequences using LIS ($n \log n$ Complexity)
 $\{3, 1, 4, 7, 6, 8, 9, 10, 6\}$

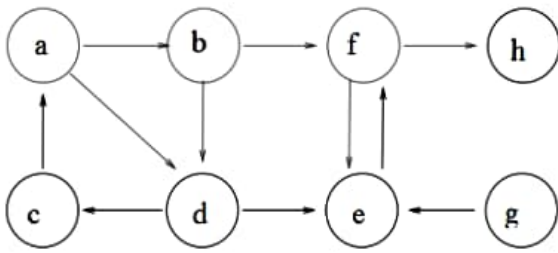
- 4) Selection of $n=6$ items, capacity of knapsack $M=10$. Determine Max Profit using 0/1 Knapsack Algorithm.

Item	Value	Weight
1	20	1
2	5	2
3	10	3
4	40	8
5	15	7
6	25	4

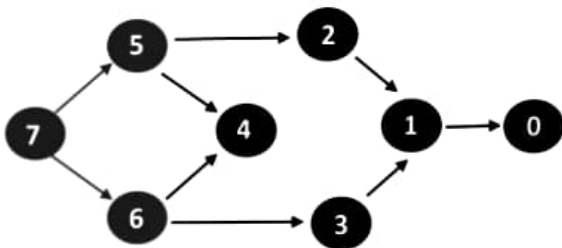
- 5) Suppose you are given an array, $A[1..N] = \{3, 2, 1, 4\}$ containing $N = 4$ positive integer numbers. You have to produce a K as a subset sum of these numbers. Solve the problem using dynamic programming.

$N = 4$, $A[1..N] = \{3, 2, 1, 4\}$, $K = 8$.

- 1) Identify different types of edges (forward edge, back edge, tree edge, cross edge, etc) using DFS graph search algorithm. Consider **a** as source. Also find the SCC in the given graph.



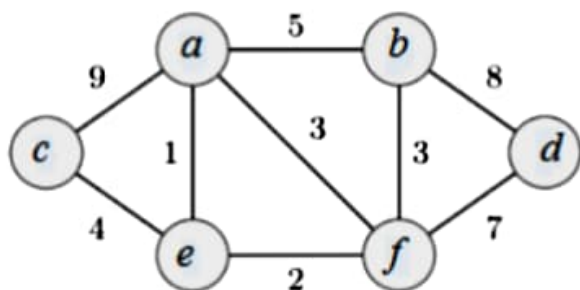
- 2) Find the topological ordering for the given graph.



Name: _____

Roll: _____

- 3) Find minimum spanning tree for the given graph using DSU data structure.



- 4) Find single source shortest path for the following graph using dijkstra algorithm Where, Source = a.

